

ing telecom era would give each user the satisfaction of getting seemingly unlimited data rates, while resources would be juggled and shared in the background. This would be achieved by predicting user demand, encouraging apps to perform bandwidth-intensive tasks during non-peak time, optimising network response times based on latency and making better use of available wireless networks.

Virtualisation at work. We are likely to see network virtualisation and cloud-like concepts implemented on the radio network. 5G is likely to use cognitive radio technology to enable different radio technologies to share the same spectrum. Base stations would be classified into radio units and baseband units, so that baseband units can be pooled to handle a high number of radio units. Optimisation of radio resource management (RRM) techniques is one of the key research areas of 5G.

Likewise, there will be a clear split between control and user planes, to facilitate heterogeneous network deployments. All user devices can be controlled on a macro layer, while user data is independently provided through a femto cell.

Smartantennae. Technologies like massive multiple-input-multiple-output (MIMO) and beam-forming active antennae will be at the heart of 5G. Smartantennae will help alter beam direction, enabling more direct communication. This would increase overall cell capacity, reduce interference and compensate for path loss with higher antenna gains. Such adaptive beam-forming algorithms are likely to be implemented on all user devices.

Versatile and responsible terminals. 5G, being a user-centric rather than operator-centric network, will shift the onus to the terminals to provide the best experience. 5G terminals are likely to have modulation schemes, error-control schemes and software-defined radios that can be downloaded from the Internet. These terminals will be able to access, choose and combine different wireless technologies and ser-



Evolution of mobile communications
(Courtesy: European Commission)

vices from different mobile/wireless access network providers.

Apart from these, research is also on to reduce battery consumption of network devices (increasing their life on the scale of months or years, rather than days or weeks), optimising deployment and operating costs, and so on.

User-centric business models

Given that 5G is disruptively different from its predecessors, many new and user-centric business models are expected to arise out of it.

User can give and take. A recent report by The Telegraph, the UK, predicts that by 2020, each person will have an average of 27 devices. These devices might be in their cars, homes, offices or bags. But just imagine having to pay for the data transacted wirelessly by all these devices? Fortunately, this scenario of service providers and users playing a clear supply and consumption role is likely to change. Like in the case of a smart power grid, where users can sell excess energy from alternative sources to the power utility, 5G—likely to be the grandfather of all networks—will allow users to make money by charging others for the coverage they provide with Wi-Fi routers or femto-cell home base stations.

No more tech pushing. The user is expected to shape the very funda-

mentals of the 5G business. Rather than being just another telecom generation, where operators push a new technology to users, it is believed that 5G will be a network that evolves to meet the demands of users from totally new markets.

Linda K. Moore, a specialist in telecommunications policy at the Congressional Research Service, writes in a recent report that, 5G might be shaped not by carriers pushing new technology

but by demand from new customers, customers that are the mainstay of important industrial sectors other than telecommunications. These include transportation, agriculture, energy, mining, defence and public safety, entertainment, medicine and finance, to cite some examples where new technologies are influencing whole industries. These industries might prefer not to manage important parts of their businesses based on somebody else's business model and may demand new techniques and service providers.

Information reselling. Since much more data is going to be generated by all our devices and apps in the 5G era, a whole industry is going to revolve around managing, mining and monetising this Big Data, albeit with privacy regulations in place.

User focus being a rather new paradigm in telecom networks, several consortiums are being formed around the world to understand the true 5G and the business models it is likely to give birth to.

All-new apps

What would you do with a mountain of money? Well, that is how you would feel when granted a seemingly unlimited data rate network connection, right? Add to this the fact that 5G is expected to have an extremely low latency, which means you would get responses instantaneously with